An interface framework and adequacy standard when generative turbulence models hit the wall

Supplementary Information

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Supplementary Note 1 Model selection and data contact

The cube data are chronologically separated from training, but the final evaluation interval is not an untouched confirmatory test. The initial 70%-training/three-integral-time-gap proposal stopped at preflight because it left 38 evaluation fields; no training or arm output was produced. The revised allocation—660 training fields, a 98-field gap and a 343-field chronological remainder—was frozen before any aligned/absent/far-time output. M0 then produced the first arm comparison on 160 evenly spaced fields from that remainder.

After M0 inspection, M1 changed capacity, training budget and sampler steps. After M1 inspection, M2 changed ordinary convolutional padding on the periodic axes. Both reused the same 160 fields. The farther-from-wall energy-score endpoint was registered before its additional ensemble inference, but after M2 and its checkpoint had been selected. The evidential labels used throughout the paper are therefore: M0, M1 and M2 are three sequential configurations of one and the same three-dimensional U-Net architecture; they differ in size, training budget, sampler length and convolutional padding, not in kind. Their defining parameters and, more importantly, their evidential status are:

- **M0** — width 24, 2,213,139 parameters, 30,000 updates, 16 sampler steps. The first-contact comparison, produced after the pre-output allocation freeze, and the only configuration whose contact with the evaluation interval preceded any arm output.
- **M1** — width 48, 8,647,587 parameters, 90,000 updates, 32 sampler steps. A model-development sensitivity: it reuses the interval M0 already contacted.
- **M2** — M1's architecture and budgets exactly, with circular streamwise and spanwise padding substituted for ordinary padding. Also a development sensitivity on the same interval, and the configuration selected for the ensemble endpoint.
- **Energy-score analysis** — a post-selection registered endpoint on the same targets, frozen after M2 and its checkpoint had been chosen.
- There is **no independent untouched final test** on this record.

Supplementary Table 1: **Evidence and data-contact ledger.** Each row states the data contact, the executed outcome and the claim boundary. "Conditional interval" elsewhere denotes a moving-block resampling interval conditional on the finite record and fitted model.

Evidence	Data contact	Executed outcome	Claim boundary
Cube M0	First arm output after allocation freeze	Complete and near aligned-band skill positive; aligned beats absent and far-time; farther-region absolute skill negative (geometric $d/h > 0.5$ label)	First-contact aggregate, not independently replayable below the terminal aggregate; 343-output count ratio $2.800T_{\text{int}}$, elapsed duration $2.792T_{\text{int}}$
Cube M1–M2	Same interval reused during capacity and padding development	Complete/near ordering retained; farther aligned-minus-absent result is configuration-sensitive	Development sensitivity, not independent confirmation
Cube ensemble	Endpoint registered after M2 selection on the same targets	Energy score favours aligned; both valid physical-statistic families fail; $M = 8$ plug-in 10–90% interval hit fraction is 51.4% but is not a calibrated finite-ensemble test	Post-selection endpoint conditional on contacted data
Candidate offline composition	Source-misidentified Case1 diagnostic; runtime software check; archived grouped-time experiment	Frozen closure export is target-member-specific and the local extractor changes calibrated features; Case1 geometry, field identity and dimensional wall quantities are unsupported; the source-valid archived scalar-stress periodic-hill study is adverse and its oracle/interface adequacy fails	Case1 withdrawn from physical interpretation. Superseded for the periodic hill by the grouped-protocol re-test row below and Supplementary Note 10
Cube direct traction	Prospectively frozen protocol on the contacted record; 240 held-out targets	Oracle first-cell traction improves every scoring region (complete gain +0.07415) and transmits 0.6311 of the matched oracle-band ceiling; absolute skill stays negative	Oracle traction bounds transmissible information; not a closure evaluation
Cube closure composition	Closure fitted on the training window; registered rule on 103 prospectively unscored frames (48 previously contacted)	Closure-predicted traction improves the source-excluded field by +0.06064; far-time traction is worse than absence; implied wall-force error falls from 0.0994 to 0.0609	Offline a-priori composition; conditional intervals; not solver-coupled

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Evidence	Data contact	Executed outcome	Claim boundary
Hill grouped re-test	Identical pipeline re-run under flattened and grouped protocols	Withdrawn gain reproduced under the flattened split (+0.73038) and reversed under the grouped split (−0.10858): a protocol artefact of 0.839	Under the grouped split, diffusion's oracle-control gain is +0.05794 [+0.00870, +0.10674]; its closure-arm gain is +0.06223 [+0.01722, +0.10573]; the flow-matching cell fails the same control and adjudicates nothing. Positive but indicative ($N_{\mathrm{eff}} \simeq 3.3$)
Reversed-time re-test under the oracle positive control	Closure and generators refitted on frames 420–1100; 25 never-tested frames from 0–299	Oracle positive control passes in both families; near-wall closure gain +0.103 with positive absolute skill; paired difference positive in 25 of 25 frames	$N_{\mathrm{eff}} = 1.21$; per-frame unanimity is the robustness evidence
Channel single-slot	Two-stage preregistration frozen before contact with a never-scored 46-frame window	Closure gains +0.120 against +0.165 for the record's own traction, with positive absolute skill; wrong-instant and shuffled tractions are adverse	Registered outcome: the interface transmits traction information; margins frozen; offline
Companion closure	Separate companion study	Closure-side geometry-transfer evidence; public citation pending author confirmation	Does not establish generator-side propagation

Every computational stage and result of this study — the sequential cube models, the registered ensemble endpoint, the traction-interface and closure-composition experiments, and the archived withdrawn diagnostics — is pinned by immutable cryptographic hashes. The complete machine-readable custody ledger, including exact file identities and the reproduction entry point, accompanies the released source-data package.

Supplementary Note 2 Aligned cube simulation record

The exact production case is included in the released code package. It uses NekRS on a $2h \times 4h \times 2h$ periodic aligned-cube domain with 14,176 conforming hexahedral elements, polynomial order 7 (7,258,112 Gauss–Lobatto–Legendre, GLL, points), order-9 cubature, TOMBO2 velocity–pressure splitting, two operator-integration-factor splitting (OIFS) substeps, target Courant–Friedrichs–Lewy (CFL) number 2 and maximum $\Delta t = 1.5 \times 10^{-3}$. Density is 1, viscosity is 1/5000, bulk streamwise velocity is held at 1, and floor, cube and ceiling faces are no-slip. High-pass filtering removes one mode with coefficient 10.

The mesh file, production configuration, terminal record and native y^+ audits are each hash-pinned in the released provenance ledger, and the mesh regenerates deterministically from the included generator. The primary production record ends at $t\overline{U}/h = 370.0005$. A separate continuation supplied the two native-audit fields at 390.0003 and 420.0001. On classified GLL wall faces, pooled maxima are 0.31631 on exposed floor, 1.33684 on cube tops and 1.92126 on vertical faces, and native wall-node velocity is zero. The audit does not classify the no-slip ceiling, although the conditioning band includes it; no ceiling-specific y^+ conclusion is drawn. The support band on the $48 \times 96 \times 48$ model grid contains 14,008 of 207,360 fluid cells. Its numerical distance threshold is $2.01(4/96)h = 0.08375h$, selecting a nominal two-cell thickness $2(4/96)h = 0.08333h$; 193,352 cells remain in the complete score. The two continuation fields are separated by $29.9998h/\overline{U}$. This exceeds the historical signal-specific integral-time estimate $29.2776h/\overline{U}$ used when the continuation was designed, but not the later conservative $36.7445h/\overline{U}$ convention used in this paper. They are therefore two native-resolution checks, not independent temporal replicates. No experimental mean/drag/Reynolds-stress comparison or grid/filter-sensitivity study is included; the aligned-cube large-eddy simulation (LES) is a numerical reconstruction reference rather than asserted truth.

Supplementary Note 3 Sequential point estimates

Model	Quantity	Complete	Near	Farther
M0	aligned R^2	+0.10308	+0.24311	−0.04030
	aligned − absent	+0.17548	+0.31980	+0.02774
	aligned − far time	+0.30688	+0.56362	+0.04408
M1	aligned R^2	+0.08949	+0.24156	−0.06623
	aligned − absent	+0.17385	+0.33198	+0.01196
	aligned − far time	+0.41305	+0.73709	+0.08136
M2	aligned R^2	+0.12049	+0.28386	−0.04678
	aligned − absent	+0.19741	+0.37017	+0.02056
	aligned − far time	+0.43420	+0.76974	+0.09074

Supplementary Table 2: **Absolute and paired point estimates across sequential model development.** Complete and near ordering is retained. Farther-from-wall absolute skill is negative in all three models. The M1 farther-region aligned-minus-absent interval crosses zero.

Only M0 terminal aggregates and historical 49-sample-block limits survive; the per-time sums of squared errors (SSE) and total sums of squares (SST) needed to replay or reblock those limits do not. We therefore do not promote them as conservative intervals. M0 member-level predictions also do not survive, so its arithmetic mean of eight generated samples per target is conditional on the realised sampler draws and its Monte Carlo error cannot be replayed. M2 retains lower-level arrays. Recomputing with a 57-evaluation-sample block gives

$$\begin{array}{lccc} & \text{complete} & \text{near} & \text{farther from wall} \\ \text{aligned} - \text{absent} & [0.18219, 0.20906] & [0.35568, 0.38195] & [0.00599, 0.03674] \\ \text{aligned} - \text{far time} & [0.40878, 0.45541] & [0.70542, 0.82500] & [0.06829, 0.11225]. \end{array}$$

These are conditional sensitivities on contacted data, not population-confidence intervals or simultaneous multiplicity-controlled statements.

Ceiling cell accounting and leak-free re-scoring. Supplementary Table 3 gives the full accounting of the computational-ceiling defect and the leak-free physical-wall re-scoring referenced in the main text. The re-scored arms use model M1, the configuration whose per-target statistics are retained.

Supplementary Note 4 Distributional and physical diagnostics

For the geometrically defined $d/h > 0.5$ vector y, ensemble members x_m, dimension D and $M = 8$, the fair off-diagonal energy-score (ES) U-statistic for a fitted continuous generator is

$$\mathrm{ES} = \frac{1}{M}\sum_{m}\frac{\|x_m - y\|_2}{\sqrt{D}} - \frac{1}{2M(M-1)}\sum_{m\neq m'}\frac{\|x_m - x_{m'}\|_2}{\sqrt{D}}.$$

Quantity	Value
Ceiling cells in the supplied band	4,608 (32.90% of the band)
identically zero (raster row 95)	2,304
bitwise duplicates of the first scored row	2,304 (1.19% of U)
Scored cells on duplicated rows	34.56% of U; 21.55% of U_{near}; 42.77% of U_{far}
Raster-unique region U_{uniq}	126,536 cells
Leak-free aligned−absent, complete	$+0.10153$ $[0.09264, 0.10904]$
Leak-free aligned−absent, near wall	$+0.21151$ $[0.20193, 0.22244]$
Leak-free aligned−absent, raster-unique	$+0.13178$ $[0.12237, 0.13976]$
Ceiling share of the published complete gain	$\approx 42\%$ of $+0.17385$

Supplementary Table 3: **Computational-ceiling accounting and leak-free physical-wall band re-scoring (M1).** Intervals are conservative block-57 moving-block intervals.

Aligned, absent-band and far-time energy scores are 0.52235, 0.52734 and 0.54838. The paired control-minus-aligned results are

$$\Delta_{\text{no}} = +0.00499\ [+0.00168, +0.00919], \qquad \Delta_{\text{far}} = +0.02603\ [+0.02091, +0.03144].$$

The intervals use the conservative 57-sample block. The original evaluation used $2M^2$; retained truth and pairwise distances permit this correction without new inference. The endpoint preserves its ordering conditionally on the selected model and contacted targets.

Physical corroboration remains null. The log root-mean-square error (RMSE) of the component spectrum does not separate aligned from far time. Nor does the normalised RMSE (NRMSE) of the standardised, instantaneous, plane-averaged quadratic fluctuation profile. The originally registered coverage-error family is invalid: 0.8 is not the finite-eight-member expected hit fraction of linearly interpolated 0.1/0.9 sample quantiles. Retained arrays do not contain ranks or member fields, so a valid rank diagnostic cannot be reconstructed. Aligned-arm descriptive hit fraction is 51.4% $[49.9\%, 52.8\%]$ under the conservative 57-sample block-selection sensitivity, but it is withdrawn from calibration and pass/fail interpretation.

Supplementary Note 5 Dependence and comparator controls

The far-time condition uses a frozen circular shift of 80 evaluation positions; paired indices differ by 172 or 173 native snapshots. It preserves a realistic wall-band source while breaking contemporaneous alignment by approximately $1.40T_{\text{int}}$. The generator was trained with condition dropout, representing the absent-band arm, but not on temporally inconsistent bands. Far-time-worse-than-absent is therefore descriptive and not a registered mechanistic signature.

We use

$$T_{\text{int}}/\Delta t = 1 + 2\sum_{k=1}^{k_0-1} \rho(k),$$

where k_0 is the first non-positive autocorrelation lag. Across band-mean energy, band-mean streamwise velocity, first-cell shear proxy, farther-plane streamwise velocity and volume turbulent kinetic energy (TKE), the largest value is $T_{\text{int}} = 122.4817$ native snapshots. The 98 omitted fields have count ratio $0.800T_{\text{int}}$, while the first evaluated state is 99 output spacings ($0.808T_{\text{int}}$) after the last training state. The 343-output evaluation remainder has count ratio $2.800T_{\text{int}}$ and elapsed duration $2.792T_{\text{int}}$; the post-spin-up record has count ratio 8.99. The conservative block is 57 evaluation samples. Four thousand circular moving-block replicates use shared indices across arms. Resampling preserves dependence and pairing but cannot create new physical events or account for model selection, multiplicity across contrasts, or between-seed variation. Earlier pipeline metadata retains a signal-specific integral-time estimate of 97.6 snapshots and an associated "one integral-correlation-time gap" label; these are historical custody metadata, not the interpretation used here. A released supersession record marks the conservative 122.4817-snapshot maximum, the 0.800-scale gap, the 2.800-scale evaluation span and the 8.99 post-spin-up event count as authoritative.

A nominal equal-support interior-shell comparison is excluded from scientific interpretation. That arm observes 13,984 cells in a shell at $0.44 < d/h \leq 0.5233$, but the evaluation scores all of those observed cells because its score mask removes only the near-wall band. The overlap is 13,984 cells in the complete score, including 9,112 cells (12.17%) of the near score and 4,872 cells (4.11%) of the farther score. The near-wall arm, by contrast, excludes all of its observed cells. The asymmetric support makes the reported interior ordering invalid as a location comparator. The terminal files remain hash-pinned for forensic custody, but no active claim uses their interior-arm values.

Frozen train-band retrieval control. Using the preregistered all-component band distance, the nearest of the 660 training bands supplies an unchanged full-field prediction for each evaluation target. Complete, near and farther fluctuation R^2 are -0.80255, -0.73251 and -0.87430. The corresponding conservative block-57 intervals are

$$[-0.84981, -0.76374], \quad [-0.78768, -0.68447], \quad [-0.91831, -0.83874]$$

for the complete, near and farther regions, respectively. The nearest donor is at least 108 native snapshots earlier than its target (median 483). The control rejects literal one-nearest-record retrieval but not more flexible memorisation or residual dependence.

Supplementary Note 6 Generality and capacity checks

All planes sharing one periodic-hill physical time remain together in train, validation or test. The diffusion checkpoint has aligned-band fluctuation $R^2 = -0.08405$ and a positive aligned-minus-absent interval; a separately parameterised flow-matching checkpoint has aligned-band $R^2 = -0.48472$ and a negative aligned-minus-absent interval (Supplementary Table 4). The test contains only $N_{\text{eff}} = 3.19$ estimated events. The M1-to-M2 padding sensitivity on the already-contacted cube interval, which is not an equivariance or general topology result, is tabulated in Supplementary Note 3.

Family	Aligned R^2	Absent R^2	Aligned − absent		Aligned − random	
Diffusion	−0.08405	−0.22606	+0.14184	[+0.09421, +0.20137]	+0.11798	[+0.05969, +0.18406]
Flow matching	−0.48472	−0.16739	−0.31693	[−0.40740, −0.23418]	−0.12856	[−0.21932, −0.04047]

Supplementary Table 4: **Physical-time-grouped hill support test.** Arms are inference-time wall-band swaps of one checkpoint per family. Intervals are 95% circular moving-block intervals (block 116, $B = 4000$, $N_{\text{eff}} = 3.19$).

Widths 48, 96 and 120 were trained for 90,000 updates under seeds 1234, 2234 and 3234 (Supplementary Tables 5 and 6). Aligned conditioning beats both controls at every width, but convergence parity, positive largest-width absolute skill and the positive size slope all fail, so no scaling claim is made.

Supplementary Note 7 Frozen closure interface custody

The closure used by the candidate adapter is not a four-input pointwise regression. Its calibrated map is

$$\widehat{C}_f = \langle T_\theta(\mathbf{\Phi}), B_\theta(\mathbf{S}) \rangle + b,$$

Width	Aligned	Absent	Far time
48	-0.03416 $[-0.05987, -0.01005]$	-0.21282 $[-0.27561, -0.15521]$	-0.13293 $[-0.17392, -0.09858]$
96	-0.04939 $[-0.08658, -0.02177]$	-0.20759 $[-0.26708, -0.14887]$	-0.17986 $[-0.20605, -0.15754]$
120	-0.07127 $[-0.11239, -0.04276]$	-0.23594 $[-0.29918, -0.17293]$	-0.20874 $[-0.24000, -0.17919]$

Supplementary Table 5: **Fixed-budget capacity matrix: absolute fluctuation R^2 by arm.** Across-seed means with crossed seed $\times$ time 95% intervals (block 116, 10,000 replicates); widths 48, 96 and 120 have 4,783,538, 18,524,258 and 28,765,946 parameters.

Width	Aligned $-$ absent	Aligned $-$ far time
48	$+0.17880$ $[+0.12713, +0.23391]$	$+0.09870$ $[+0.07195, +0.13164]$
96	$+0.15821$ $[+0.08731, +0.23178]$	$+0.12997$ $[+0.09414, +0.17312]$
120	$+0.16401$ $[+0.08385, +0.24529]$	$+0.13691$ $[+0.10100, +0.17120]$
Capacity slope	-0.00989 $[-0.03928, +0.02067]$	$+0.02173$ $[+0.00633, +0.03563]$

Supplementary Table 6: **Fixed-budget capacity matrix: paired contrasts and capacity slopes.** Slopes are in R^2 per natural-log parameter under the crossed resampling of Supplementary Table 5; the aligned-arm slope itself is -0.01855 $[-0.03451, -0.00030]$.

where $\mathbf{\Phi} = (U_m/U_e, y_m/\delta_{99}, \delta_{99}\partial_x p/U_e^2, \delta^*_{\mathrm{res}}\partial_x p/U_e^2)$ is a local station row. The permutation-invariant branch receives a member-level set with the same four columns plus β_p. The exported inference header has a 4–64–64–16 trunk, a fixed 16-component branch embedding and ten descriptor knots. It does not contain learned branch-network weights or set normalisation. Source inspection traces the embedding to the $Re = 10{,}595$ periodic-hill target member's own ten-station, wall-stress-free set (Krank *et al.*, Flow, Turbulence and Combustion **101**, 521–551, 2018). The header exactly replays that member when given its calibration-consistent local rows (maximum C_f discrepancy 2.55×10^{-10}), but it cannot construct $B_\theta(\mathbf{S})$ for a new source. Here p is kinematic pressure, and the fifth set feature is

$$\beta_p = \frac{\delta^*_{\mathrm{res}}\,\partial_x p}{u_p^2}, \qquad u_p = (\nu|\partial_x p|)^{1/3}.$$

The displacement thickness uses the same resolved support $y_m \le y \le \delta_{99}$ as the fourth feature. The magnitude $|\partial_x p|$ defines the non-negative velocity scale, the sign is carried by $\partial_x p$, and the primary implementation assigns $\beta_p = 0$ at zero pressure gradient. Thus the missing branch depends on a fully specified signed, dimensionless coordinate, not an undefined label.

The archived companion tree also retains deterministic training code and a standardised station bundle containing the local rows, padded member sets, masks and training targets. It retains no full-model checkpoint: the public headers store only a target-member embedding and trunk. The architecture and fit can therefore be regenerated only by performing a new empirical training run; they cannot be replayed as the frozen closure used for an inference-only E2 evaluation.

The available local adapter also changes the feature contract. Calibration uses a source-native differentiated wall-pressure trace and integrates displacement thickness only over $y_m \le y \le \delta_{99}$. The adapter instead uses five-station-smoothed $-U_e\partial_x U_e$ and integrates from the wall to δ_{99}. Table 7 reports a deterministic audit on the ten retained calibration-member profiles. It performs no fitting, generator inference or transfer evaluation.

This audit yields two independent hard boundaries. Reusing the exported embedding on a new flow is a target-member transplant, not source-general closure inference; recomputing an embedding requires newly fitted branch weights, the archived standardisation contract and a prospectively defined source-valid station set. And even on the calibration member, replacing native pressure gradient and resolved displacement thickness materially changes the signed output. Thus no closure-conditioned launch is admissible until the exact feature definitions, branch construction, normalisation, source identity and information-matched comparator are frozen. The

Interface diagnostic	Value
Pressure surrogate relative RMSE / Pearson correlation	0.961/0.098
Full/resolved displacement-thickness ratio, median [range]	1.345 [1.292, 2.366]
Closure-output relative RMSE / sign changes	0.408/1
Calibration-consistent / substituted-feature C_f R^2	0.558/0.171
Frozen family-operator separated-subset C_f R^2	−2.037
Paired local-row multilayer-perceptron (MLP) separated-subset C_f R^2	+0.722

Supplementary Table 7: **Frozen closure interface-custody audit.** The first four rows measure the effect of changing feature definitions on the retained Krank calibration member. They are interface-sensitivity diagnostics, not transfer estimates. The local-row multilayer perceptron (MLP) has no family/set input and is therefore an adverse secondary baseline, not an information-matched comparison with the family-conditioned operator.

compact replay record and per-source hashes accompany the released source data.

Supplementary Note 8 Withdrawn Case1 diagnostic

Source-native identity audit. The candidate E2 study is withdrawn from physical interpretation. Its raw arrays are the public CoNFiLD Case1 record: coordinates have shape $(918, 2)$ and data have shape $(16001, 918, 3)$. Du *et al.* (Nature Communications **15**, 10416, 2024) identify the case as a two-dimensional irregular pipe with stochastic forcing and the fields as (u, v, p). Their Supplementary Information specifies 918 mesh cells, time step 0.05 and forward-Euler integration. The local rasterizer instead described the third component as w (although it retained only the first two), labelled the geometry a wall jet or backward expansion and searched an alpha-shape raster for an embedded obstacle. It found none and substituted $h = L_y/6$. The downstream evaluator imposed $Re_h = 2800$ and inferred $\nu = U_b h/Re_h$. Native wall identities, wall normals, mesh connectivity, step height and viscosity needed for a signed stress or skin-friction interpretation are absent from the downloaded arrays and cited numerical description.

The resulting δ_{99} and displacement thickness are columnwise constructions, while $\partial_x p = -U_e \partial_x U_e$ replaces the available pressure channel. The claimed reference stress is a first-cell velocity-gradient proxy using the imposed viscosity. Consequently, correlation, sign agreement, C_f, separation and cross-kind transfer from this construction have no source-valid physical meaning. Every raw and derived artefact of this pipeline is hash-pinned in the released custody ledger.

Data contact and estimator audit. The rasterizer computed channel means and standard deviations over all 16,001 frames before selecting every fourth frame and making the chronological split. The conditional generators were then fitted on this same record with instantaneous target-derived first-cell proxy stress and 30% condition dropout. In evaluation, the closure feature is static because it is derived from the target time mean, whereas the random control spatially permutes the instantaneous target proxy. The controls therefore do not all match temporal information. Each family has one fitted training realisation. The reported cellwise empirical-distribution continuous ranked probability score (CRPS) pools standardised components, locations and times for a 16-member fitted ensemble; it does not include model-fit uncertainty and does not demonstrate calibration. These defects remain even apart from the fatal source identity.

The following tables preserve the computed values to prevent selective reporting. They are forensic outputs of the unsupported construction, not estimates for a wall jet, separated flow or closure-to-generator system. Total-field paired means use the original 36-snapshot circular resampling block.

A separately archived runtime software check instantiates the software path from one spatially coarsened snapshot through the frozen closure and direct-stress adapter into both frozen generator

Family	Contrast	Mean [95% interval]
Diffusion	closure − absent band	$+0.01491$ [$+0.00555, +0.02455$]
Diffusion	closure − random	-0.00890 [$-0.02310, +0.00511$]
Diffusion	closure − wrong mean	-0.01365 [$-0.02560, -0.00154$]
Flow matching	closure − absent band	$+0.01425$ [$+0.00407, +0.02428$]
Flow matching	closure − random	$+0.06370$ [$+0.04429, +0.08249$]
Flow matching	closure − wrong mean	$+0.02385$ [$-0.00098, +0.04918$]

Supplementary Table 8: **Withdrawn Case1 scalar-feature outputs.** The values are retained unchanged, but they support no physical composition claim because the source identity and dimensional wall quantities are invalid. The controls are also not temporally information-matched.

Diagnostic	Diffusion	Flow matching
Mean-field R^2	0.9374	0.9640
$u'u'$ profile R^2	0.4458	0.1267
$v'v'$ profile R^2	-0.6672	-0.7839
Closure absolute fluctuation R^2	-0.2194	-0.1989
Closure − absent band, near total R^2	-0.0198	-0.0720
Closure − absent band, farther total R^2	$+0.0192$	$+0.0295$

Supplementary Table 9: **Unfavourable outputs retained from the source-invalid Case1 pipeline.** "Near" and "farther" inherit the unsupported $h = L_y/6$ construction and therefore have no physical wall-normal interpretation.

families. It is a central-processing-unit (CPU)-only check with six targets, $N_{\text{eff}} = 0.167$, two ensemble members and six sampler steps. Its block limits collapse to the same value and its prospective verification criteria reject the single-snapshot proxy inputs, target-member-dependent branch and inadequate sample/ensemble sizes. It is retained as software commissioning evidence, not as causal E2 performance evidence, and it does not repair the Case1 source identity.

Archived grouped-time adequacy failure. A larger archived E2 artefact uses the independently identified periodic-hill record and keeps the three spanwise slices of each physical hill time in the same partition: training times 1–479 and test times 590–959. It evaluates 370 test times ($N_{\text{eff}} = 3.246$), with eight conditional members, 32 sampler steps and a 114-time circular block. Supplementary Table 10 reports the unaltered adverse endpoints.

Family	Score	Closure history	Absent feature	Oracle history
Diffusion	total-field R^2	$+0.53084$	$+0.50569$	$+0.52658$
	fluctuation R^2	-0.03792	-0.09355	-0.04735
Flow matching	total-field R^2	$+0.39887$	$+0.51184$	$+0.45226$
	fluctuation R^2	-0.32986	-0.07994	-0.21175

Supplementary Table 10: **Archived scalar-stress experiment retained as an adequacy failure.** The diffusion closure-minus-absence total-field contrast is $+0.02480\,[+0.00380, +0.04308]$; the flow-matching contrast is $-0.11334\,[-0.13715, -0.08519]$. All fluctuation scores are negative, and the flow-matching oracle arm is worse than absence. Intervals are conditional 95% circular moving-block intervals with a 114-time block and shared indices across arms.

This periodic-hill artefact cannot decide a causal runtime composition claim. Its reference "stress" is the scalar first-cell proxy $\nu u(y = \Delta y/2)/(\Delta y/2)$, while the runtime input is a stride-two subsample of the target field rather than an independent coarse-solver state. The fitted generator was trained on oracle/current/empty scalar features and then exposed to closure outputs with relative stress RMSE 0.841. For the cube surface, a valid interface instead requires both signed wall-on-fluid tangential traction components in a declared local frame, with wall normals, tangent bases, normalisation and lift support. The archived training code also seeds minibatch draws but not network weight initialisation before model construction; it is one training realisation per family.

The result, code, checkpoint and input-data hashes are pinned in the released custody ledger, and the exact result file is released with the source data. It is retained to prevent selective reporting, not promoted as positive or negative evidence about a scientifically adequate closure-to-generator composition.

Supplementary Note 9 Quarantined evidence

The legacy 527-frame periodic-hill split distributes spanwise planes from the same physical time across training and test. Every test frame therefore has a same-time counterpart in training, and a nearest-neighbour audit gives fluctuation $R^2 = 0.982$. All downstream flattened-split hill scores, reach, calibration, spectral, rough-wall and adversarial derivatives are excluded from the active manuscript and Supplementary displays. The underlying files are preserved in explicitly named quarantine directories for forensic traceability.

The active source package also exposes the complete periodic-hill re-test set, under distinct names and without overwriting any frozen artefact: the direct-traction and band-lift grouped runs, the 64×192 port control and the 96×288 raster sensitivity, the stochastic-sampler arm with its pre-launch protocol, amendment, sampler unit test and mechanically applied decision rule, and the single interval estimator behind every hill number. The flow-matching cell's failed positive control under both samplers and both rasters, the refutation of its preregistered explanation, the hill closure arm's near-wall interval spanning zero and its non-separation from a scrambled traction all remain in the active package.

The active source package exposes the cube point and ensemble artefacts, M0 terminal aggregate, archived linear/deterministic controls, compact cube LES provenance, the withdrawn Case1 ledgers, physical-time-grouped hill summary, negative capacity diagnostic, archived E2 adequacy failure, and the single-slot channel package (two-stage preregistration, frozen decision applicator, block-inference calibration and mechanically computed outcome). No unfavourable outcome is removed: null physical corroboration, negative farther-from-wall absolute skill, failed grouped-hill replication, failed scaling, the source-invalid Case1 output and the contradictory periodic-hill E2 family outcomes remain in the active package.

Supplementary Note 10 Grouped-protocol hill re-test

The grouped re-test (Methods) re-executes the frozen closure→traction→generator composition on the separating periodic-hill record under both the flattened array-index split and the leakage-free physical-time-grouped split, with matched architecture, budget, seeds and sampler. A training-free nearest-neighbour retrieval null scores +0.983 under the flattened split, with 88% of retrieved neighbours being same-time twins (twin fluctuation correlation 0.977–0.996), against −0.862 under the grouped split with none. The closure itself predicts held-out traction with $R^2 = 0.757$ against 0.557 for the equilibrium law, the Reichardt band lift reproduces the true band with $R^2 = 0.871$, and the diffusion sampler repair improves the development-window absent branch from −0.469 at 32 steps to −0.212 at 256.

Two conditioning representations. In the *direct* representation the generator receives the surface traction itself; its oracle reference arm is the record's own exact direct numerical simulation (DNS) traction. In the *band lift* that traction is first expanded into a near-wall velocity band through the closure's Reichardt profile; its oracle arm is the record's exact DNS band velocities and therefore has lift fidelity $R^2 = 1$ by construction. The value $R^2 = 0.871$ (0.915 at 96×288) is the fidelity of the *closure* lift, not of the oracle arm, and the equilibrium lift attains essentially the same band fidelity (0.874 and 0.915): band-velocity fidelity does not by itself distinguish the closure from the equilibrium law.

One named interval estimator. Every hill interval quoted in this paper is the hierarchical bootstrap Methods declares—seeds resampled with replacement, then a circular moving-block resample in physical time with block 123—recomputed from the retained per-frame components and released with the source data. The earlier released audit averaged the per-frame sum of squared errors over the realised seeds before resampling time, which conditions on the sampler draws and gives narrower intervals; both are reported for every arm. Adopting the declared estimator changes the zero-exclusion verdict of 7 of the 180 arm/region/cell contrasts, in every case from excluding zero to spanning it, and in five of the seven the affected arm is a *control* rather than a method arm. No headline hill result changes. Supplementary Table 11 collects the registered endpoints under this estimator.

Resolution sensitivity of the grouped protocol. The leakage-free grouped protocol had only ever been executed at 64×192, and the finer 96×288 raster only under the flattened split, so the combination that decides the question was completed here. The raster is the single varied factor; the per-wall-face closure embedding makes its parameter shape raster-dependent, so the closure was refitted by a verbatim port of the same procedure (same seeds, optimiser, normalisation and by-physical-time fit/validation split), and the sampler budget was re-selected by the same frozen rule scoring the baseline arm only on the development window. Split contract, arms, mixture, architecture, capacity, training steps, members, metrics, regions and block bootstrap are unchanged. A control run of the same ported script at 64×192 reproduces the released result on every quantity that matters (H2 oracle $+0.0565$ against the released $+0.0569$; H1 oracle -0.0621 against -0.0593; calibration $0.369 \rightarrow 0.534$ against $0.371 \rightarrow 0.537$), confirming the instrument before any difference is attributed to resolution. The reading was fixed in advance: a raster whose oracle band does not clear zero reports nothing for that cell, and the coarse result is not superseded.

Prospective test of the under-dispersion explanation (stochastic sampler). The flow-matching cell's failure was initially attributed to ensemble under-dispersion. That attribution was tested rather than left standing. Family F's released sampler is the zero-diffusion member of the stochastic-interpolant family sharing the same marginals (Albergo, Boffi and Vanden-Eijnden 2023; Ma *et al.* 2024, both cited in the main text): with $\hat{z} = x_t + (1-t)v$ the interpolant's noise estimate and $\Delta = t - t_n$, the step is $x_{t_n} = x_t - \Delta v - \Delta\lambda\hat{z} + \sqrt{2\lambda t\Delta}\,\xi$, which at $\lambda = 0$ is algebraically the released Euler step. Every generator weight, the closure, the 32-step budget, the split, the six arms, the eight members, the three seeds, the regions, the block bootstrap and the decision rule were inherited unchanged; nothing was retrained. The single coefficient λ was selected on the development window, scoring only the unconditioned arm, by minimising the distance of its rank-histogram extreme-bin mass to the calibrated 2/9—a two-sided criterion that penalises over-dispersion equally and that returns $\lambda = 0$ (the null repair) if no positive value is better calibrated. The test window was contacted once, at the selected $\lambda = 8$, together with a $\lambda = 0$ port-fidelity control.

Before launch, the two candidate stochastic samplers were checked on an analytically solvable Gaussian problem whose exact interpolant velocity is known in closed form. The construction first proposed—the denoising-diffusion implicit model (DDIM) η step transported to the linear interpolant—was rejected there: it does not preserve the terminal marginal under exact prediction and *reduces* ensemble spread for an under-dispersed model ($\times 0.71$), so it could not have tested the hypothesis. The stochastic-interpolant step passes all four checks and was substituted before any outcome on this record was observed.

Ensemble dispersion, measured uniformly and used as a diagnostic. An earlier draft raised ensemble calibration to a standing adequacy criterion beside the positive control. That is withdrawn. That criterion's justification was the hypothesis that under-dispersion causes the flow-matching cell's failure, and the stochastic-sampler experiment above tested that hypothesis prospectively and refuted it. Supplementary Table 14 therefore applies one executable measurement to every hill cell and reports it as a diagnostic.

The statistic is the fraction of rank-histogram mass in the two extreme bins of the nine-member histogram, whose calibrated value is $2/9 = 0.222$; the interval is the same hierarchical seed-then-block bootstrap used everywhere else. The rule, applied identically to every cell, is that a cell is *consistent with* uniformity when its interval covers 2/9. With $N_{\text{eff}} \simeq 3.3$ this test has little power, so covering uniformity is not evidence of calibration—only a rejection is informative, and that asymmetry is stated wherever the number appears.

Read uniformly, the measurement does two things. It shows a real association at 64×192: every flow-matching cell rejects uniformity, strongly under-dispersed, and every one fails its positive control, while every diffusion cell at that raster is consistent with uniformity and passes. And it shows why that association cannot be promoted to an acceptance criterion: the 96×288 diffusion cell rejects uniformity in the *opposite*, over-dispersed direction (0.111) and nevertheless passes its positive control more strongly than any other cell. A criterion that would have discounted the paper's strongest hill cell is not a criterion.

Supplementary Note 11 Reversed-time cube re-test under an oracle positive control

The reversed-time re-test refits the closure and four generators from scratch on frames 420–1100 of the cube record and scores 25 frames drawn from 0–299, a window that had never been a test unit, 132 frames from training against a measured integral time $\tau = 97.6$. Both generative families are trained at an identical budget with a single frozen conditioning mixture that includes the oracle band, the near-wall region $d \leq 0.5$ is co-primary with the whole field, and no interface conclusion is drawn unless the oracle band beats absence with an interval excluding zero (the oracle positive control). All numbers in Supplementary Table 15 were frozen as estimands before any outcome existed.

Quantity	Flow matching (F)	Diffusion (G)
Direct-traction representation, grouped split, complete scored volume		
Oracle DNS traction − absent	−0.05022 [−0.20713, +0.09563]	+0.05794 [+0.00870, +0.10674]
Closure traction − absent	−0.10971 [−0.27105, +0.03696]	**+0.06223 [+0.01722, +0.10573]**
Equilibrium traction − absent	−0.05386 [−0.13278, +0.03600]	+0.05169 [+0.03077, +0.07631]
Wrong-instant traction − absent	−0.29246 [−0.34783, −0.21921]	+0.01676 [−0.00099, +0.03476]
Scrambled traction − absent	−0.18540 [−0.25996, −0.10550]	+0.03639 [+0.02422, +0.04567]
Paired method-vs-control contrasts, recomputed arm-minus-arm per replicate		
Closure − oracle traction	−0.05949 [−0.09754, −0.02791]	+0.00429 [−0.00651, +0.01694]
Closure − equilibrium	−0.05585 [−0.14616, +0.02761]	+0.01055 [−0.01474, +0.03138]
Closure − wrong instant	+0.18276 [+0.04048, +0.31537]	+0.04547 [+0.00117, +0.08643]
Closure − scrambled	+0.07570 [−0.15774, +0.29069]	+0.02584 [−0.02051, +0.06724]
Regional, absolute and distributional		
Closure − absent, near the wall	−0.06663 [−0.31674, +0.14541]	+0.06299 [−0.02181, +0.15166]
Closure − absent, outer region	−0.12145 [−0.25965, +0.01103]	+0.06204 [+0.02700, +0.09432]
Absolute R^2, absent arm	−0.06773	−0.20962
Closure − absent, CRPS	+0.05325 [+0.03408, +0.07335]	−0.01241 [−0.01882, −0.00558]
Closure − absent, energy score	+0.08800 [+0.05974, +0.11741]	−0.01717 [−0.02678, −0.00755]
Band-lift representation: traction expanded through the Reichardt profile		
Oracle DNS band − absent, 64×192	−0.05927 [−0.14218, +0.01304]	+0.05689 [+0.01147, +0.10740]
Closure band − absent, 64×192	−0.18325 [−0.30402, −0.08949]	+0.03844 [−0.01689, +0.09129]
Oracle DNS band − absent, 96×288	+0.02490 [−0.07225, +0.10577]	+0.07919 [+0.02749, +0.13055]
Closure band − absent, 96×288	−0.04515 [−0.15243, +0.05109]	+0.05872 [−0.01111, +0.12215]
Lift fidelity, closure / equilibrium	0.871/0.874 (64×192); 0.915/0.915 (96×288)	
Lift fidelity, oracle band	1.000: it *is* the record's own band	
Protocol and other endpoints		
Closure − absent, flattened split	+0.73038 [+0.70156, +0.75094]	—
Held-out spanwise-plane robustness	−0.0837	+0.0494
Wall-force correlation, absent → closure	0.036 → 0.542	−0.104 → 0.445
Wall-force relative RMSE, absent → closure	1.987 → 1.318	2.082 → 1.852

Supplementary Table 11: **Grouped-protocol re-test endpoints, one named estimator throughout.** Source-excluded fluctuation-R^2 contrasts against each cell's own absent branch under the grouped physical-time protocol unless labelled otherwise. Intervals are the hierarchical seed-then-moving-block bootstrap Methods declares ($B = 4000$, block 123); proper scores are lower-is-better. The load-bearing result is the diffusion cell's direct-traction closure arm, with its limits in the same table: indistinguishable from the exact traction and from the equilibrium law, separated from a wrong-instant but not a scrambled traction. With $N_{\text{eff}} \simeq 3.3$ every entry is indicative. Dashes denote arms not run in that family.

Oracle band − absent	Flow matching	Diffusion
64×192 (decisive run), whole field	-0.059 $[-0.142, +0.013]$	$\mathbf{+0.057}$ $[\mathbf{+0.011}, \mathbf{+0.107}]$
64×192 (ported control), whole field	-0.0621 $[-0.1375, +0.0002]$	$\mathbf{+0.057}$ $[\mathbf{+0.012}, \mathbf{+0.106}]$
96×288, whole field	$+0.025$ $[-0.072, +0.106]$	$\mathbf{+0.079}$ $[\mathbf{+0.027}, \mathbf{+0.131}]$
96×288, near the wall	$+0.085$ $[-0.109, +0.267]$	$\mathbf{+0.152}$ $[\mathbf{+0.040}, \mathbf{+0.277}]$
96×288, outer region	$+0.008$ $[-0.064, +0.062]$	$\mathbf{+0.058}$ $[\mathbf{+0.019}, \mathbf{+0.095}]$
96×288, closure arm, whole field	-0.045 $[-0.152, +0.051]$	$+0.059$ $[-0.011, +0.122]$
96×288, wrong-instant / scrambled, near wall	-0.263 / -0.291	-0.075 / -0.020
96×288, extreme-bin mass, absent → oracle	$0.355 \to 0.429$	$0.111 \to 0.108$

Supplementary Table 12: **The positive control across two rasters.** The diffusion cell passes at both and more strongly at the finer raster; the flow-matching cell passes at neither and does not keep the sign of its point estimate between them, which is why it adjudicates nothing in either direction. Bold marks intervals excluding zero in the direction of the control. The scrambled control is weakly positive over the whole volume in the diffusion cell ($+0.018\,[+0.004, +0.038]$), so part of that gain is non-specific; near the wall it is correctly negative.

Quantity	Deterministic ($\lambda = 0$)	Stochastic ($\lambda = 8$)
Rank-histogram extreme-bin mass, unconditioned	0.371	**0.181**
the same under oracle conditioning	0.537	0.282
calibrated reference	0.222	0.222
Positive control (oracle band − absent), whole field	-0.059 $[-0.146, +0.013]$	$\mathbf{-0.134}$ $[\mathbf{-0.190}, \mathbf{-0.084}]$
near the wall	-0.008 $[-0.219, +0.157]$	-0.085 $[-0.234, +0.036]$
outer region	-0.073 $[-0.129, -0.025]$	-0.148 $[-0.186, -0.113]$
Closure − absent, whole field	-0.183 $[-0.311, -0.090]$	-0.224 $[-0.319, -0.151]$
Wrong-instant control, near the wall	-0.363	-0.393
Scrambled control, near the wall	-0.213	-0.215
Port fidelity: worst deviation from the released run	0.0 (exact, all 18 arm/region cells)	

Supplementary Table 13: **The under-dispersion explanation, tested and refuted.** The repair restored calibration as designed—closing 72% of the distance to a calibrated ensemble with byte-identical network weights—and the oracle positive control then failed *more* clearly, its interval moving from spanning zero to strictly below it. The explanation is therefore refuted for this cell, and no mechanism is claimed. Ensemble dispersion is accordingly a diagnostic throughout, never an acceptance criterion; Supplementary Table 14 applies the same measurement to every hill cell. Controls keep their expected ordering under both samplers. The preregistration, its amendment and the decision rule are released with the source data.

Cell	Family	Extreme-bin mass: absent	Extreme-bin mass: oracle	vs 2/9	Positive control
Direct traction, 64×192	F, flow matching	0.372 $[0.357, 0.385]$	0.558 $[0.545, 0.572]$	rejects	fails
Direct traction, 64×192	G, diffusion	0.241 $[0.221, 0.258]$	0.229 $[0.210, 0.243]$	covers	passes
Band lift, 64×192 (release)	F, flow matching	0.371 $[0.357, 0.383]$	0.537 $[0.526, 0.548]$	rejects	fails
Band lift, 64×192 (release)	G, diffusion	0.248 $[0.221, 0.275]$	0.242 $[0.213, 0.267]$	covers	passes
Band lift, 64×192 (port)	F, flow matching	0.369 $[0.355, 0.381]$	0.534 $[0.524, 0.544]$	rejects	fails
Band lift, 64×192 (port)	G, diffusion	0.233 $[0.207, 0.260]$	0.229 $[0.201, 0.256]$	covers	passes
Band lift, 96×288	F, flow matching	0.355 $[0.342, 0.366]$	0.429 $[0.415, 0.441]$	rejects	fails
Band lift, 96×288	**G, diffusion**	$\mathbf{0.111}$ $[\mathbf{0.102}, \mathbf{0.121}]$	$\mathbf{0.108}$ $[\mathbf{0.099}, \mathbf{0.117}]$	**rejects**	**passes**
Band lift, stochastic sampler	F, stochastic sampler	0.181 $[0.172, 0.189]$	0.282 $[0.270, 0.296]$	rejects	fails
Band lift, deterministic control	F, port control	0.371 $[0.357, 0.383]$	0.537 $[0.525, 0.548]$	rejects	fails

Supplementary Table 14: **Ensemble dispersion applied identically to every hill cell, as a diagnostic and not an acceptance criterion.** Fraction of rank-histogram mass in the two extreme bins ($2/9 = 0.222$ when calibrated), with hierarchical seed-then-block 95% intervals. "Covers" means the interval contains $2/9$; with $N_{\text{eff}} \simeq 3.3$ this is weak evidence and only a rejection is informative. The bold row is decisive against using this statistic as an acceptance criterion: it rejects uniformity by over-dispersion yet carries the strongest passing positive control in the study.

Quantity	Value
Positive control (oracle band − absent)	$+0.130$ $[0.115, 0.145]$ flow matching; $+0.029$ $[0.026, 0.032]$ diffusion
Near-wall closure − absent (flow matching)	$+0.103$ $[0.091, 0.115]$; absolute $-0.098 \to +0.005$; 79% of the oracle-band gain
Matched controls, near-wall absolute	shuffled -0.128; wrong-instant -0.164 (both below absent)
Nested-shell closure − absent	$+0.266$ $[0.221, 0.317]$ ($d \leq 0.15$); $+0.103$ ($d \leq 0.5$); $+0.053$ $[0.046, 0.063]$ (whole field); $+0.006$ $[0.004, 0.008]$ ($d > 0.5$)
Nearest scorable shell, absolute	$+0.149$ against -0.117 with nothing supplied
Per-frame paired difference	positive in 25/25 frames for both flow-matching seeds (minimum $+0.042$; medians $+0.108$, $+0.096$); diffusion 25/25 and 24/25
Wall-force endpoint	correlation $-0.14 \to +0.51$; relative RMSE $0.158 \to 0.125$; shuffled control correlation $+0.61$ (force integral invariant under permutation)
Registered rule	fails one clause: diffusion unconditional absolute skill -0.53 against the -0.14 floor; sampler sweep saturates at -0.792, -0.650, -0.641 over 32, 64, 128 steps
Closure traction skill (held-out)	correlation 0.852; zero-reference 0.549; centred -0.021 (equilibrium law: 0.423 zero-reference, -0.305 centred)

Supplementary Table 15: **Reversed-time re-test endpoints.** Conservative moving-block intervals conditional on the record; $N_{\text{eff}} = 1.21$ for the decisive window, which is why per-frame unanimity is reported as the primary robustness evidence.

Supplementary Note 12 Representative prior work

Supplementary Table 16 places the present interface study beside representative field-generation and wall-observation work. The comparison is deliberately qualitative: the inputs, regimes and targets differ, so the rows identify what wall information enters each model and whether an explicit closure-to-generator interface is evaluated.

Supplementary Table 16: **Representative prior work and the present evaluation wedge.** Rows compare the supplied information and near-wall or interface role; they do not form a numerical ranking. Full citations appear in the main-text reference list.

Work	Flow/target	Conditioning	Near-wall or interface treatment
Gao *et al.* (2023)	Backward-facing step, channel and turbulent boundary layer	Coarse flow, initial state or prescribed statistics	Explicit wall-bounded generation and super-resolution; no wall-stress closure-to-generator interface
CoNFiLD (2024)	Periodic hills, rough-wall turbulence and channel flow	Sparse sensors, coarse fields or corrupted data (zero-shot Bayesian conditioning)	Continuous field generator; near-wall discrepancies are documented, without an explicit wall closure interface
Oommen–Karniadakis (2026)	Schlieren jet super-resolution, isotropic-turbulence forecasting and cylinder-wake reconstruction	Coarse fields, flow histories or sparse measurements	Field, perceptual and adversarial objectives with spectral evaluation; no explicit wall closure or closure-to-generator composition
Cuéllar *et al.* (2024)	Three-dimensional channel volume	Wall pressure and shear	Deployable wall signals with wall-normal error analysis
Parikh *et al.* (2026)	Near-wall channel turbulence	Full, sparse, partial or low-resolution wall signals	Conditional generation with quantified uncertainty
This work	One aligned-cube LES record, one wall-resolved channel record and one separating periodic-hill record	Oracle velocity band; oracle first-cell wall traction; **closure-predicted** wall traction	Explicit closure-to-generator interface: a physics-grounded closure is fitted on the training partition and composed with the generator, scored outside every supplied and every closure-read cell. Offline a-priori only

Supplementary Note 13 Reproducibility

A single entry point in the released code package verifies the pinned source and simulation-case hashes, regenerates the production mesh deterministically, rebuilds the source data and released displays, compiles the manuscript and this Supplementary Information, and independently reconstructs the M2 point, ensemble and train-band-retrieval endpoints from lower-level sufficient statistics; a clean-room rebuild repeats the run from a manifest-only copy of the tree. M0 is checked at the terminal aggregate level only. A reviewed source-first contract checks every headline and decision-bearing claim against source values selected before comparison, and each figure-generation script separately compares the values it plots with source-derived expectations.

Training and inference are explicitly seeded and seeds are reused across matched arms, but graphics-processing-unit (GPU) library benchmarking is enabled and checkpoints omit complete random-number-generator state, so result custody and inference replay are distinguished from exact bitwise retraining or interrupted-run continuation. The raw aligned-cube volume and model checkpoints are byte-verified locally but are too large to distribute with the source-data package; they are available from the corresponding author on request during review and are deposited in a permanent public archive with a DOI on publication. The complete custody ledger — immutable hashes and file identities for every script, result, checkpoint and raw record — accompanies the released source-data package.